Customer Purchase Analysis

Part 1: DATA ANALYTICS FUNDAMENTALS

1. Types of Data Analytic:

A. Descriptive Analysis: what happened? , Understand the past.
Example: A company analyzing last month's sales.

B. Diagnostic Analysis: Why did it happen? , Find causes.
Example: A company investigating the reasons for a decline in sales.

C. Predictive Analysis: What will happen? ,Forecasts what might happen in the future. Example: A company forecasting sales for the coming month.

D. Prescriptive Analysis: What should we do?, Suggests the action that should be taken.
Example: A company decides to offer discounts to increase sales.

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2. Data Types:

A. Categorical Data: Data used to classify items into categories.
divided into:

- Nominal: Categories without a specific order.
  - Example: Car color (red, blue, black).
- Ordinal: Categories with a specific order
  - Example: Satisfaction level (low, medium, high).

B. Numerical Data: Data representing numbers.
divided into:

- Discrete: Countable numbers (integers).
  - Example: The number of students in the class (30 students).
- Continuous: Measurable data that can take on various values (fractions/decimals).
  - Example: A student's weight (58.5 kg)

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3. Data cleaning : The process of detecting errors in data and correcting or removing them to improve its quality.

   importance: It helps obtain accurate and reliable analysis results.

   common issues:

   - Missing Values : solution → Fill them with suitable values.
   - Duplicate Data : solution → Delete duplicate records.
   - Inconsistent Data: solution→ Standardizing the data format.

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4. Business Understanding :

   Problem :The company relies on discounts (promo codes) and subscriptions to incentivize customers, yet the data does not show that this leads to higher customer spending. Furthermore, women—who account for 32% of revenue—are completely excluded from these offers.

   Questions :

   1. Do customers who used the discount or have a subscription spend more than others?
   2. Who are the customers generating revenue, and who is excluded from the programs?
   3. Which segments generate the revenue, and where does the opportunity lie?

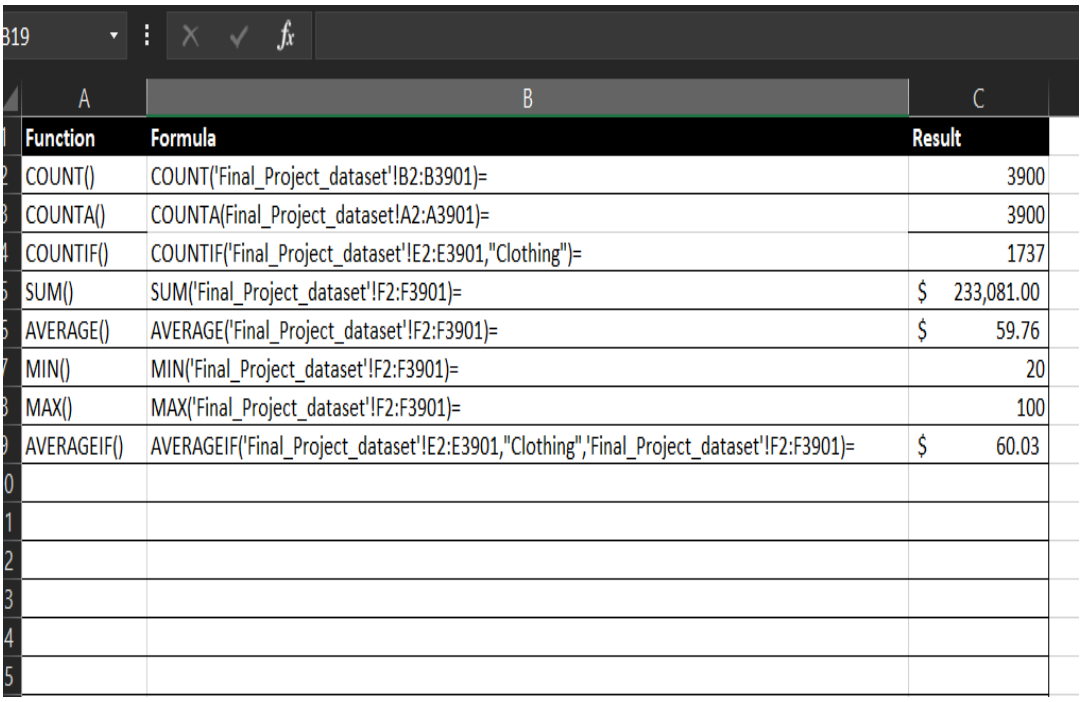
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PART 2 — EXCEL DATA ANALYSIS

Task 1 Excel Functions :

- COUNT() : It counts the number of cells that contain numbers.
- COUNTA() : Counts the number of non-empty cells, whether they contain numbers or text.

- COUNTIF() : It counts the cells that meet a specific condition.
- SUM() : Summing numerical values.
- AVERAGE() : Calculating the arithmetic mean.
- MIN() : Determine the smallest value in the column.
- MAX() : It determines the maximum value in the column.
- AVERAGEIF() : It calculates the average of values that satisfy a specific condition.



	A	B	C
1	**Function**	**Formula**	**Result**
2	COUNT()	COUNT('Final_Project_dataset'!B2:B3901)=	3900
3	COUNTA()	COUNTA(Final_Project_dataset!A2:A3901)=	3900
4	COUNTIF()	COUNTIF('Final_Project_dataset'!E2:E3901,"Clothing")=	1737
5	SUM()	SUM('Final_Project_dataset'!F2:F3901)=	$ 233,081.00
6	AVERAGE()	AVERAGE('Final_Project_dataset'!F2:F3901)=	$ 59.76
7	MIN()	MIN('Final_Project_dataset'!F2:F3901)=	20
8	MAX()	MAX('Final_Project_dataset'!F2:F3901)=	100
9	AVERAGEIF()	AVERAGEIF('Final_Project_dataset'!E2:E3901,"Clothing",'Final_Project_dataset'!F2:F3901)=	$ 60.03
10			
11			
12			
13			
14			
15			

Task 2. Pivot Tables:

Pivot Table 1: Total Purchase Amount by Category

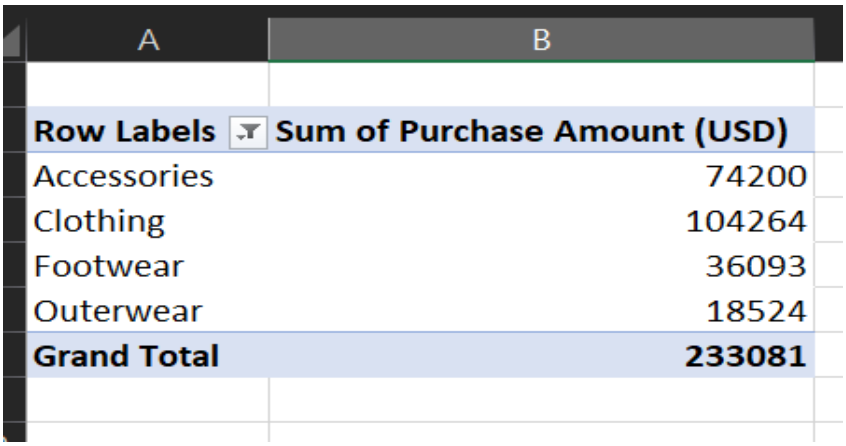
Question : What is the total value of purchases for each category?

Pivot Table Configuration:

- **Rows:** Category
- **Columns:** None

- **Values:** Purchase Amount (USD)
- **Aggregation:** SUM

Explanation: "Category" was placed in the Rows area and "Purchase Amount (USD)" in the Values area, using the SUM function to calculate the total for each category. The results showed that the "Clothing" category recorded the highest total ($104,264), followed by "Accessories" ($74,200), "Footwear" ($36,093), and "Outerwear" ($18,524), bringing the grand total to $233,081.



Row Labels	Sum of Purchase Amount (USD)
Accessories	74200
Clothing	104264
Footwear	36093
Outerwear	18524
Grand Total	**233081**

Pivot Table 2: Average Purchase Amount by Season and Category

Question : How does the average purchase value vary by season and category?

Pivot Table Configuration:

- Rows: Season
- Columns: Category
- Values: Purchase Amount (USD)
- Aggregation: AVERAGE

Explanation: "Season" was placed in the rows area and "Category" in the columns area to compare purchases by season and category, while "Purchase Value (USD)" was placed in the values area using the AVERAGE function. The highest average was recorded for the "Footwear" category during fall ($63.71), whereas the lowest average was recorded for the "Outerwear" category during spring ($54.63). Overall, the "Footwear" category recorded the

highest average purchase value ($60.26), while the "Outerwear" category recorded the lowest ($57.17).

A	B	C	D	E	F
Average of Purchase Amount (USD)	**Column Labels**				
Row Labels	**Accessories**	**Clothing**	**Footwear**	**Outerwear**	**Grand Total**
Fall	61.33950617	61.40515222	63.71323529	59.76136364	61.55692308
Spring	56.50166113	60.99559471	58.6196319	54.62962963	58.73773774
Summer	60.98717949	56.56372549	58.70625	57.04	58.4052356
Winter	60.36633663	60.87946429	60.57142857	57.025	60.35736354
Grand Total	**59.83870968**	**60.02533103**	**60.25542571**	**57.17283951**	**59.76435897**

Task 3. Excel Charts:

Chart 1: Number of Purchases by Payment Method

Question: Which payment method is used most frequently?

Explanation: A bar chart was created to compare the number of purchases made using each payment method. This chart helps identify the most frequently used payment method and provides insight into customer preferences regarding payment options.

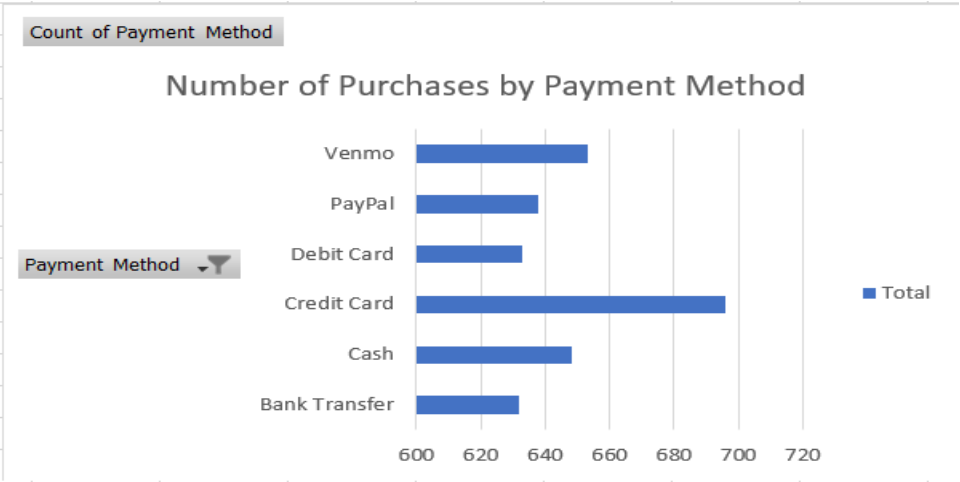
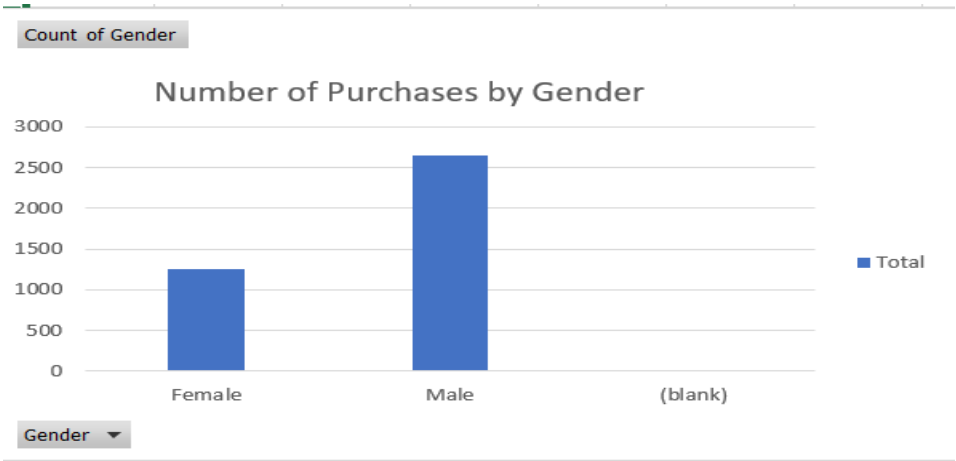


Chart 2: Number of Purchases by Gender

Question: How does the number of purchases differ between male and female customers?

Explanation : A bar chart was created to compare the number of purchases made by male and female customers. The chart reveals differences in purchase frequency between the two groups.



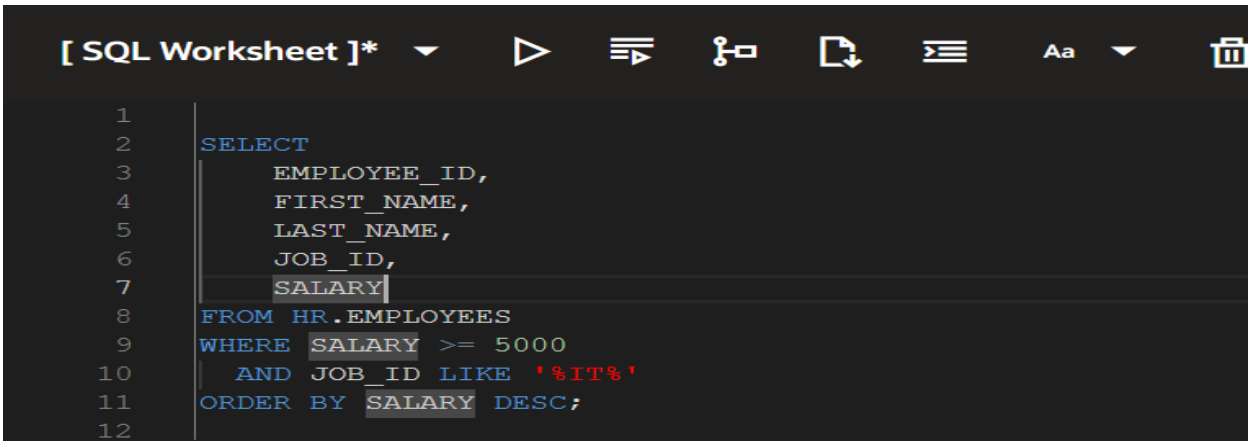
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PART 3 — SQL DATA ANALYSIS

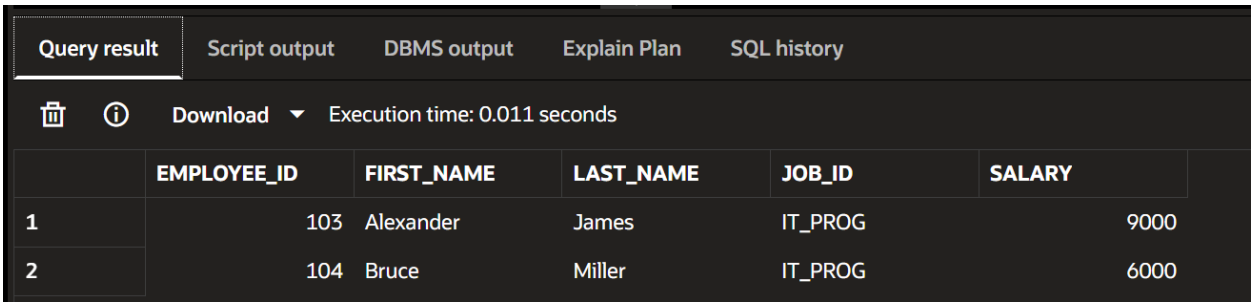
Query 1 — Filtering and Sorting:

Question: Which employees earn a salary of at least $5,000 and work in an IT-related job?

Code:



Result:



Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.011 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	JOB_ID	SALARY
1	103	Alexander	James	IT_PROG	9000
2	104	Bruce	Miller	IT_PROG	6000

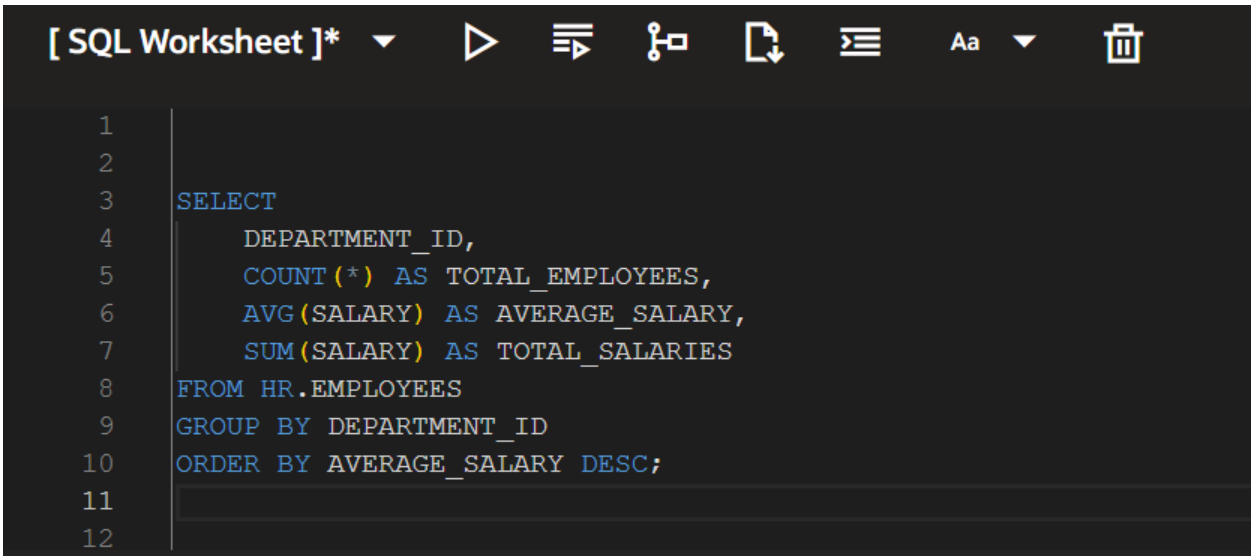
Query Usage: This SELECT query retrieves employee data from the HR.EMPLOYEES table for employees earning a salary of 5,000 or more whose JOB_ID contains the characters "IT" (using the WHERE and AND clauses), and then sorts the results by salary in descending order (using ORDER BY SALARY DESC).

*Note: The query identifies employees who meet the salary and job code criteria, displaying the highest salary at the top of the results.

Query 2. Aggregation and Grouping :

Question: What is the number of employees and the average salary in each department?

Code:



[SQL Worksheet]*

```
1
2
3   SELECT
4       DEPARTMENT_ID,
5       COUNT(*) AS TOTAL_EMPLOYEES,
6       AVG(SALARY) AS AVERAGE_SALARY,
7       SUM(SALARY) AS TOTAL_SALARIES
8   FROM HR.EMPLOYEES
9   GROUP BY DEPARTMENT_ID
10  ORDER BY AVERAGE_SALARY DESC;
11
12
```

Result:

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.008 seconds

	DEPARTMENT_ID	TOTAL_EMPLOYEE	AVERAGE_SALARY	TOTAL_SALARIES
1	90	3	19333.333333333332	58000
2	110	2	10154	20308

Query Usage: This SQL query is used to analyze employee salaries by department; it displays the department ID and the number of employees in each department, while also calculating the average and total salaries. The query groups employees by department ID using `GROUP BY DEPARTMENT_ID` and sorts the departments in descending order based on the average salary using `ORDER BY AVERAGE_SALARY DESC`, placing the department with the highest average salary at the top.

*Note: The results show the number of employees and salaries for each department. The department listed in the first row records the highest average salary within the group.

Query 3. Filtering Groups :

Question: Which departments have at least five employees, and how do their salary ranges compare?

Code:

[SQL Worksheet]*

```
3
4   SELECT
5       DEPARTMENT_ID,
6       COUNT(*) AS TOTAL_EMPLOYEES,
7       AVG(SALARY) AS AVERAGE_SALARY,
8       MIN(SALARY) AS MINIMUM_SALARY,
9       MAX(SALARY) AS MAXIMUM_SALARY
10  FROM HR.EMPLOYEES
11  GROUP BY DEPARTMENT_ID
12  HAVING COUNT(*) >= 5
13  ORDER BY TOTAL_EMPLOYEES DESC;
14
```

Result:

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.008 seconds

	DEPARTMENT_ID	TOTAL_EMPLOYEE	AVERAGE_SALARY	MINIMUM_SALARY	MAXIMUM_SALAR
1	50	45	3475.5555555555557	2100	8200
2	80	34	8955.882352941177	6100	14000

Query Usage: This query uses `GROUP BY DEPARTMENT_ID` to group employees by department, `COUNT(*)` to count the number of employees in each department, `AVG(SALARY)` to calculate the average salary, and `MIN(SALARY)` and `MAX(SALARY)` to determine the lowest and highest salaries. It also uses `HAVING COUNT(*) >= 5` to display departments with five or more employees, and finally `ORDER BY TOTAL_EMPLOYEES DESC` to sort the departments in descending order based on the number of employees, from highest to lowest.

*Note: The query returns only those departments with five or more employees. The first row represents the department with the highest number of employees among those that met the criteria.

Query 4. Your Business Question :

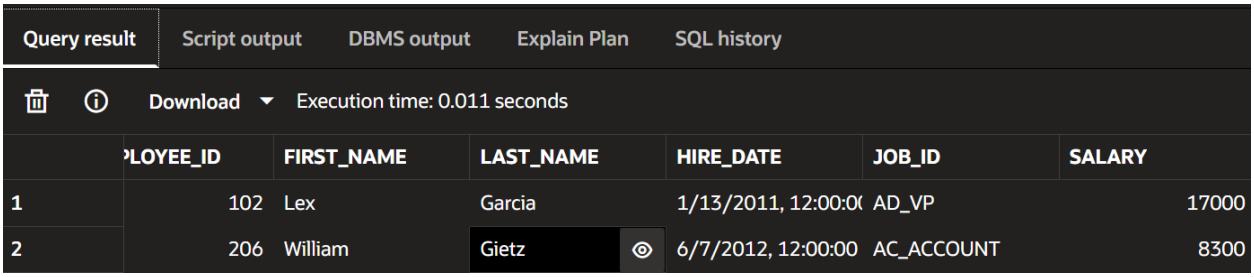
Question: Which employees were hired before January 1, 2023, and what are their job titles and salaries?

Code:

[SQL Worksheet]*

```
2
3
4   SELECT EMPLOYEE_ID,
5          FIRST_NAME,
6          LAST_NAME,
7          HIRE_DATE,
8          JOB_ID,
9          SALARY
10  FROM HR.EMPLOYEES
11  WHERE HIRE_DATE < DATE '2023-01-01'
12  ORDER BY HIRE_DATE ASC;
13
```

Result:



Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.011 seconds

	LOYEE_ID	FIRST_NAME	LAST_NAME	HIRE_DATE	JOB_ID	SALARY
1	102	Lex	Garcia	1/13/2011, 12:00:0(	AD_VP	17000
2	206	William	Gietz	6/7/2012, 12:00:00	AC_ACCOUNT	8300

Query Usage: This query displays the employee number, name, hire date, job code, and salary for employees hired before January 1, 2023. The `WHERE` clause is used to select employees who meet the date criterion, while `DATE` specifies the required date; `ORDER BY HIRE_DATE ASC` is used to sort the results from the earliest hire to the most recent.

*Note: It identifies employees hired before January 1, 2023, and lists their hiring dates in ascending order, enabling us to identify the company's longest-serving employees.

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PART 4 — PYTHON & EDA

Task 1. Load & Understand the Data :

Summary→ The dataset was loaded using the Pandas library and consists of **3,900 rows and 19 columns** containing customer demographic data and shopping information. It comprises 4 integer columns, 1 float column, and 14 categorical columns, with no missing values. The "Clothing" category appears most frequently (1,737 records), while "Male" is the most frequent gender category (2,652 records). Overall, this dataset facilitates the analysis of customer purchasing behavior and product preferences.

Task 2. Data Quality :

Summary→The dataset contains **"**no missing values or duplicate row**"** consequently, no deletion or processing was required. The data types are generally appropriate, with numerical and categorical columns correctly identified. Checks for unique values revealed diversity across categories such as items, locations, and seasons. Furthermore, conducting additional checks regarding spelling, value ranges, and logical consistency would enhance the data's reliability.

Task 3. Descriptive Statistics :

Summary→Descriptive statistics were used to summarize numerical and categorical data. The results showed that the average customer age was 44.07 years, the average purchase value was $59.76, and the average review rating was 3.75 out of 5. The total purchase value amounted to $233,081, with individual purchase values ranging from $20 to $100. These statistics provide an overview of customer characteristics and purchasing behavior.

Task 4. Exploratory Analysis :

Summary→

Analysis 1- Frequency of product categories

Clothing is the most frequent product category, accounting for 1,737 records (44.54%) of the dataset. Accessories ranks second with 1,240 records (31.79%), while Outerwear has the lowest frequency with 324 records (8.31%). This suggests that Clothing and Accessories represent the largest share of the purchase records and may deserve particular attention in inventory and marketing planning.

Analysis 2- Total purchase amount by category

Clothing has the highest total purchase amount at $104,264, followed by Accessories at $74,200. Outerwear has the lowest total at $18,524. These results show that Clothing contributes the largest share of the recorded purchase amount. Management could review product availability and marketing strategies across categories to support business performance.

Analysis 3- Average purchase amount by season and category

The seasonal analysis showed that Footwear in Fall had the highest average purchase amount at approximately $63.71, while Outerwear in Spring had the lowest average at approximately $54.63. This indicates differences in average purchase amounts across seasonal and product-category combinations. Management could use these patterns to review seasonal promotions and product planning.

Analysis 4- Average review rating by category

Footwear has the highest average review rating at 3.79, followed by Accessories at 3.77 and Outerwear at 3.75. Clothing has the lowest average rating at 3.72. The differences are small, suggesting that average customer ratings are relatively similar across categories.

Task 5. Findings :

Finding 1: Clothing is the most frequent product category

Clothing accounts for 1,737 records (44.54%), making it the most frequent category, followed by accessories with 1,240 records. This finding can assist companies in optimizing inventory planning and marketing strategies, though further analysis is required to assess profitability.

Finding 2: Higher representation of male customers

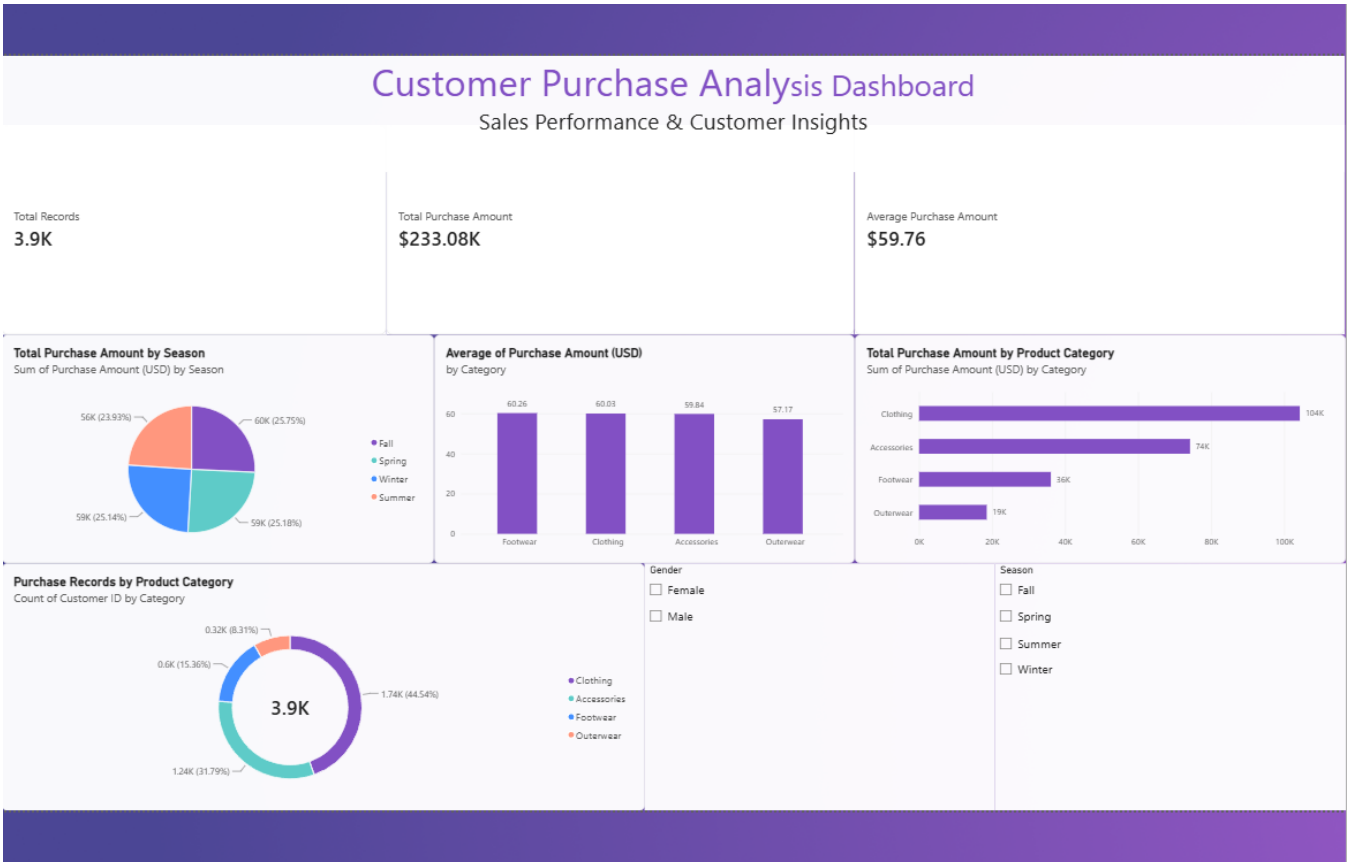
Male customers account for 2,652 records (68%), compared to 1,248 records for female customers (32%). This information can help companies understand customer demographics and evaluate marketing strategies; however, comparing spending rates is necessary to identify differences in purchasing patterns.

Finding 3: Average purchase value is $59.7

The average purchase value is $59.76, with values ranging from $20 to $100. Companies can use this average to monitor spending and analyze purchasing patterns across various categories, seasons, and discount scenarios.

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PART 5 — POWER BI DASHBOARD



Insights & Recommendations :

Insight 1: Clothing is the most frequently occurring product category.

Clothing accounts for 1,737 out of 3,900 records, representing approximately 44.54% of the dataset. The accessories category ranks second in terms of frequency, with 1,240 records. This indicates that clothing is the dominant product category in terms of purchase frequency.

Insight 2: Variation in Purchase Amounts Across Product Categories.

Total purchase amounts vary significantly across product categories; the clothing category records the highest total at $104,264, followed by accessories at $74,200, footwear at $36,093, and finally outerwear at $18,524. These results demonstrate that the categories contribute varying proportions to the total purchase value.

Insight 3: Purchasing activity is relatively consistent across seasons.

The seasonal chart shows that total purchase amounts are relatively similar during autumn, spring, and winter, while summer records a somewhat lower total. This indicates that purchasing activity is distributed throughout the year rather than being concentrated in a single season.

Recommendation 1: Optimize inventory for high-purchase categories.

Management should review inventory availability for clothing and accessory categories, given their significant share of purchase records. Inventory levels should be adjusted to align with actual demand while avoiding excessive stock accumulation.

Recommendation 2: Developing Seasonal Marketing Campaigns.

Management should examine the reasons behind the relative decline in purchase volumes during the summer and consider launching targeted promotions or campaigns for seasonal products. Furthermore, the impact of these initiatives must be evaluated by comparing purchase volumes across different seasons, both before and after the implementation of each campaign.